CC Tutorial - 2

Demonstration of a PaaS Application: Creating a Trailblazer App on Salesforce Trailhead

Observation:

Session - 2

Demonstration of PaaS Application : creating a Trailblazer app or on salesforce Trailhead.

What is PaaS ?

PaaS is a cloud service model that delivers a complete development and deployment environment in the cloud, including infrastructure, middleware, database and development tools. You manage your application and data.

Benefits:

1) Faster development : No need to configure server, built in development tools, quick deployment
2) Cost effective : Pay as you go pricing, No hardware purchase, reduced maintanance costs
3) Scalability : Automatic scaling based on traffic, handles high load easily
4). Simplified development : one click deployment, CI/CD integrations available
5). Better collaboration : integrated version control system
6). Security & updates managed : reduced operational burden

Eg : salesforce, AWS elastic beastalk, google app engine, ...

Trailhead : official online learning platform by salesforce technologies through interactive modules, hands-on exercises and real projects. Helps developers, administrators and users learn salesforce skills.

Steps to build apps using salesforce lightning :

(1). Use lightning app builder.
- drag and drop interface with features like creating pages, adding components, designing dashboards
(2). Create lightning components :
- developers build reusable UI elements called lightning components such as buttons, forms,
(3). connect to salesforce data

Setting up a trailblazer playground

Trailblazer playground is a personal salesforce developer environment provided through salesforce trailhead where users can practice, test and build applications safely
- Sign up for salesforce trailhead and launch the playground.

creating trailblazer app

- Building basic app in salesforce
  - creating custom objects
    Eg : student objects with fields like email, name, course
  - adding fields and data types.
    Eg : datatype for course is text and email is email
  - designing UI through no-code or low-code development methods

Automating processes in trailblazer app

What is process automation in salesforce ?

- Process automation in salesforce is the use of built-in tools to automatically execute business processes based on specific triggers, conditions or events.

Eg: When a new student record is created, send welcome email automatically
This saves time and manual effort, reduces human error and improves workflow efficiency.

Flow builder: used to create complex workflows using a visual interface. Automates record updates, approvals
Eg:- creating a flow for automatically sending welcome email when two new student record and created.
- defining trigger, add action, configure email, save and activate.

Process builder: Another tool to automatically perform actions when certain conditions are met. Now, flowbuilder is encouraged compared to process builder.

5: Deploying and Testing the Trailblazer app. Developers test the app to verify that automation rules, flows and components behave as expected.

Steps to test:
(1). Test the features
e.g.- Forms accept correct data, automation rules run properly, records are created or updated correctly

(2). Use debug logs:
to check for errors, flow execution.
(3) debug flow builder
(4). use developer console.

sandbox: It is a copy of salesforce production environment used for testing, development and debugging without affecting real data.

Q. How companies use salesforce PaaS for business apps?
Allows companies to build, customize and deploy business applications on the cloud without managing infrastructure.
CRM, Sales automation, customer support systems, business process automation, customer support system enterprise applications, integration with other systems

Salesforce Trailhead Demonstration:

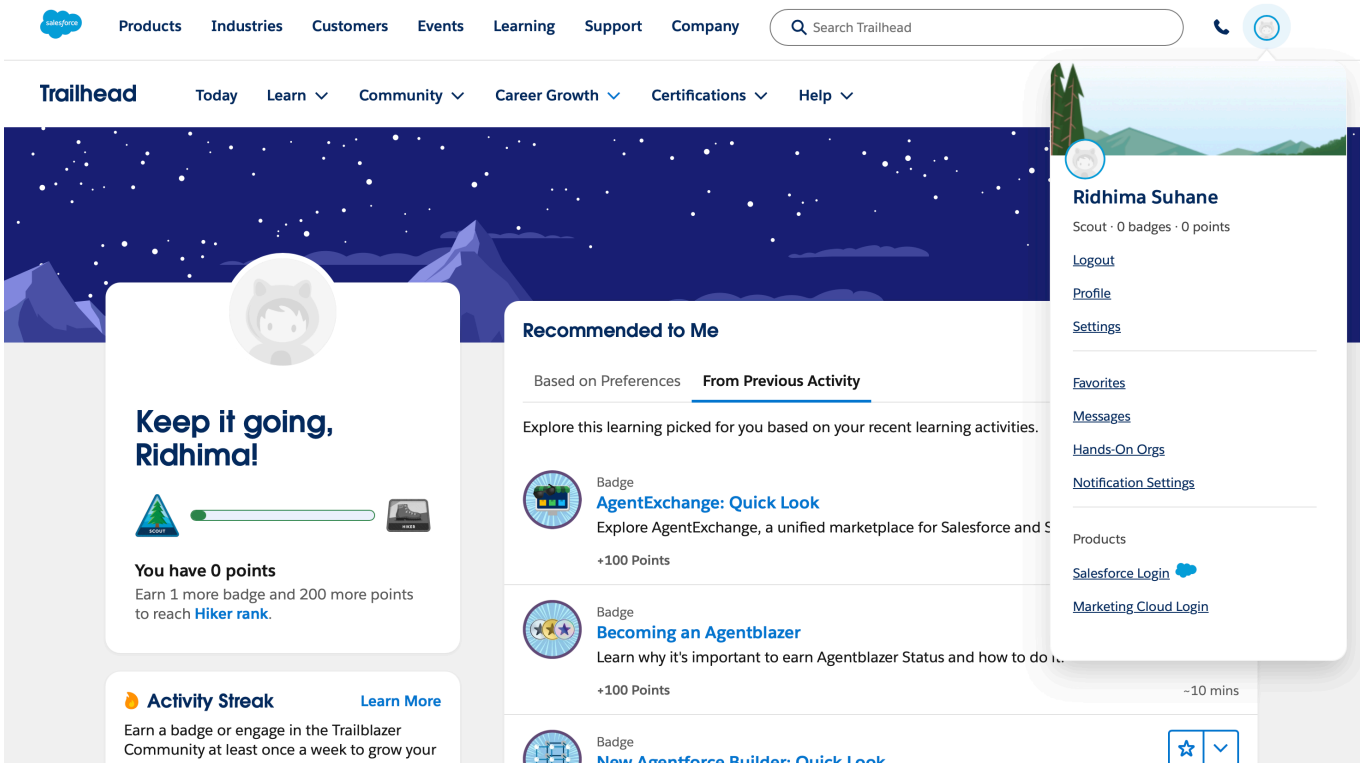

Fig 1.1: Salesforce Trailhead HomePage

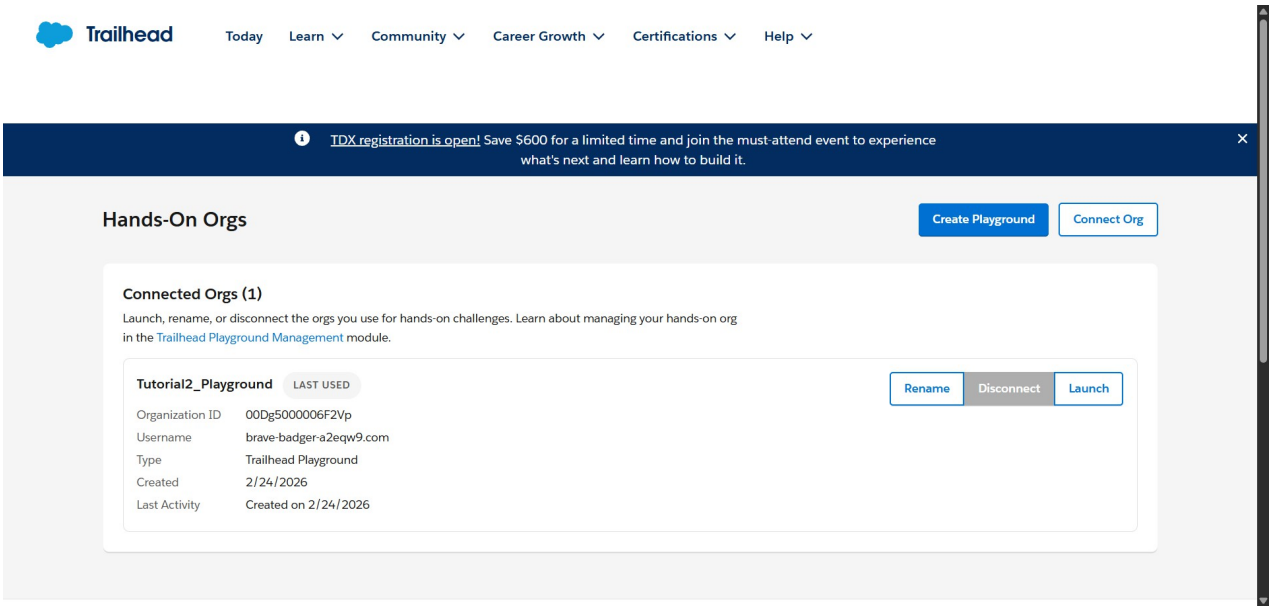

Fig 1.2: Playground creation

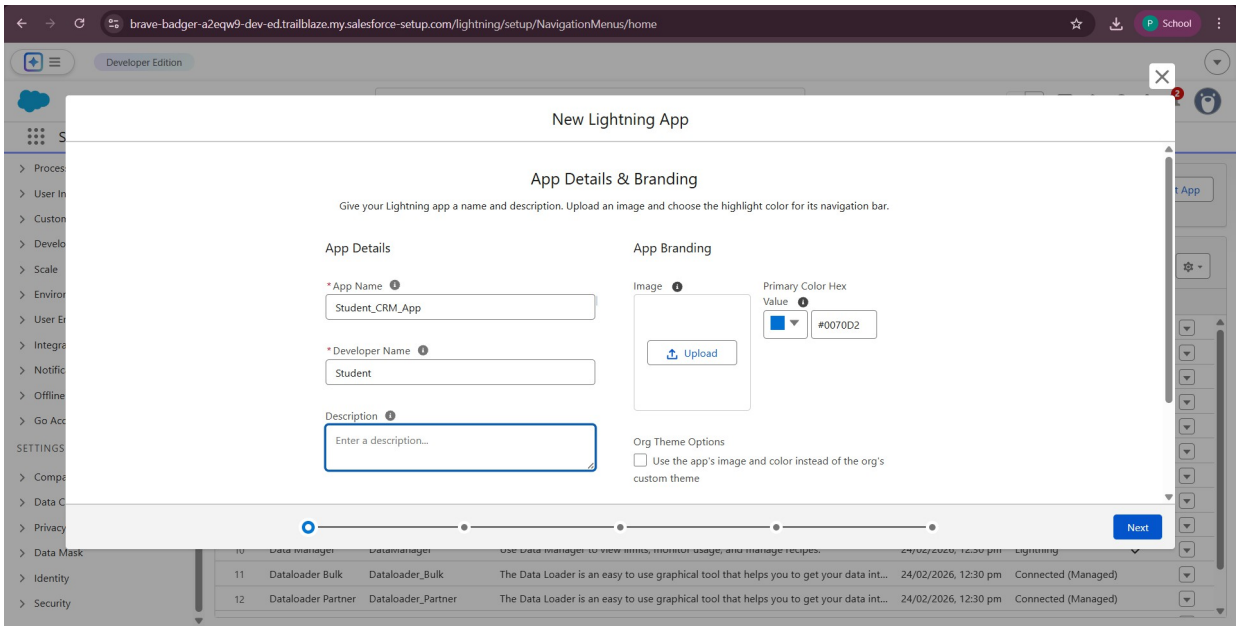

Fig 1.3: New Lightning App creation

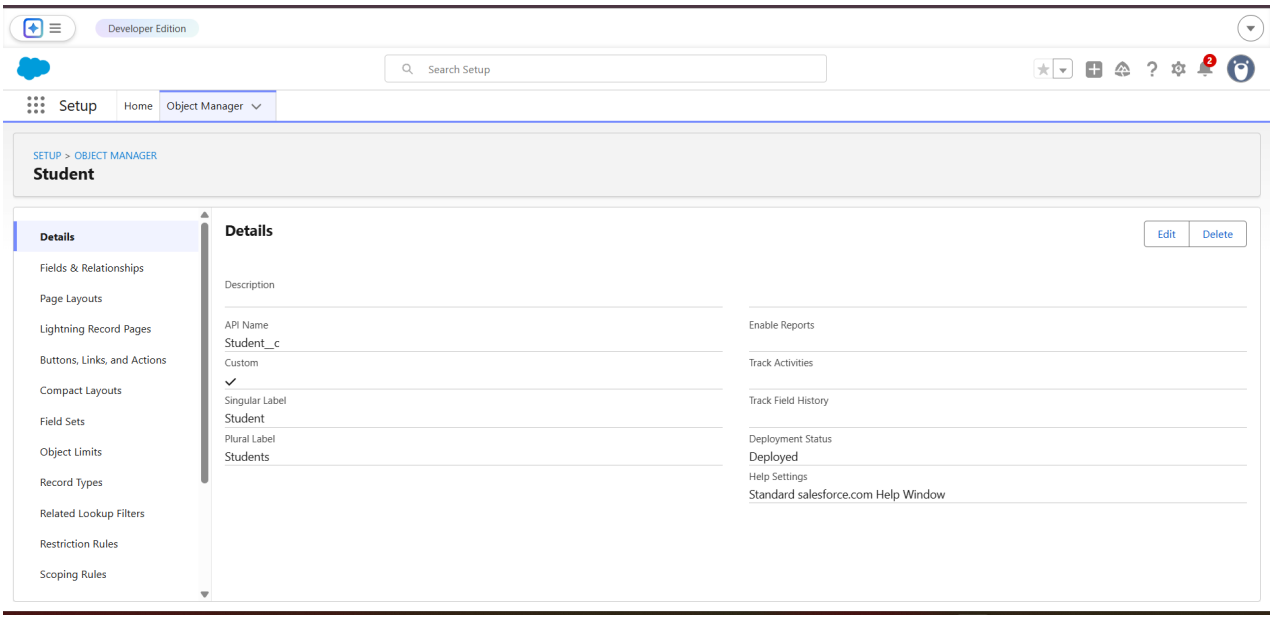

Fig 1.4: Adding a custom object - Student Object creation

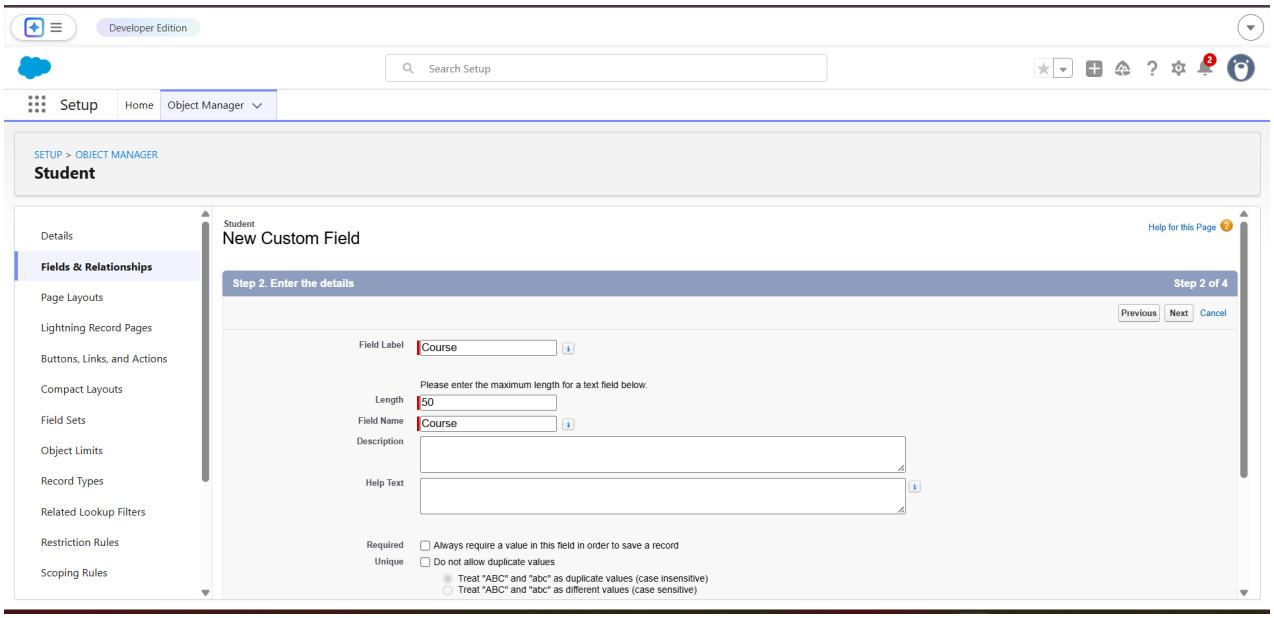

Fig 1.5: Adding a custom field

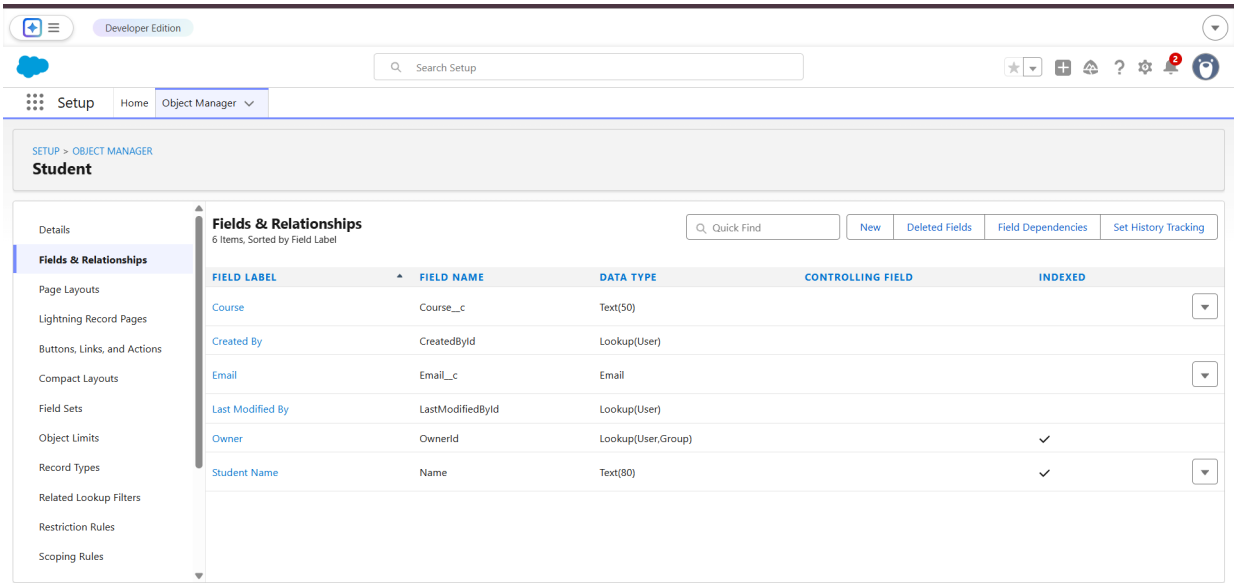

Fig 1.6: All fields and relationships

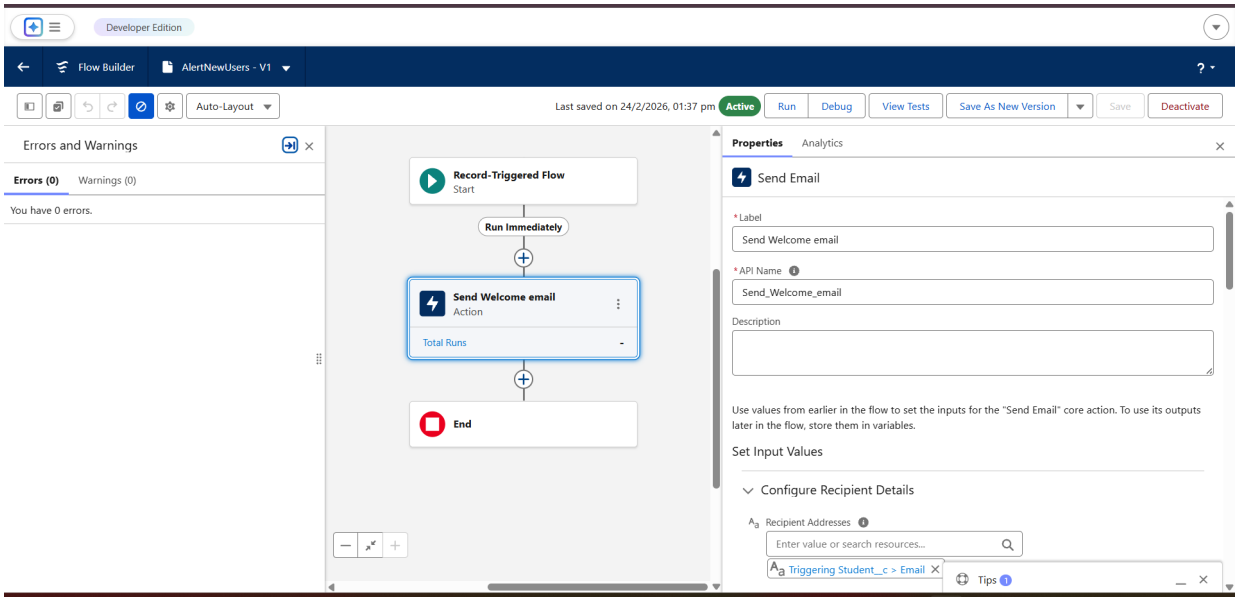

Fig 1.5: Flow Builder flow creation - setting up automation rules

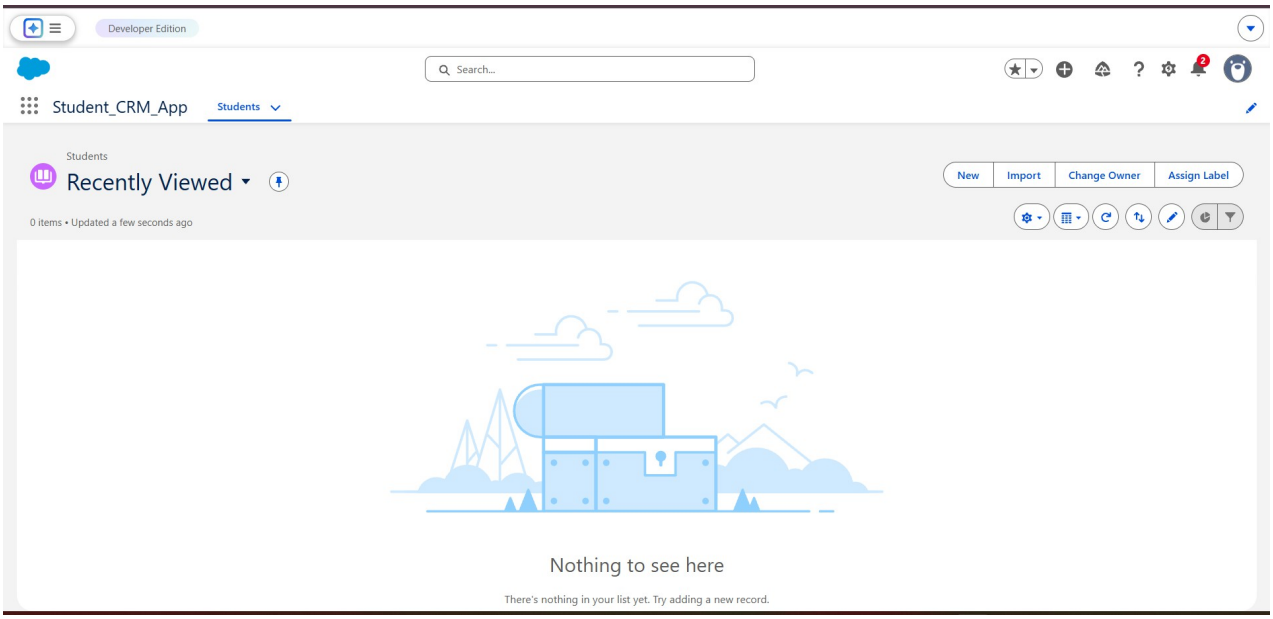

Fig 1.6: Newly created Sudent_CRM_App